Machine Learning and Data Mining (ST443)

Zoltán Szabó

LSE

Contents: warmup



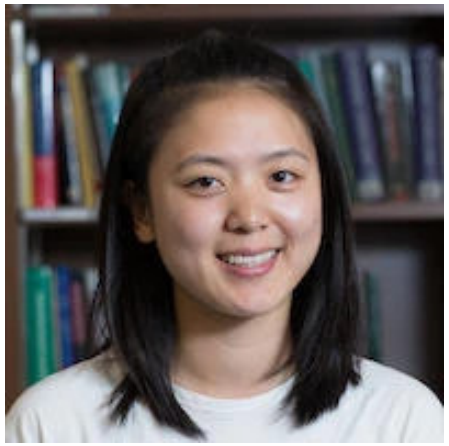
- Team.
- References & authors.
- ML hints.
- Admin: assessment, genAI policy.
- High-level idea: prototype tasks.
- (Linear) algebra: recap.

Team

Team [notation: link]

- Seminar: Phil Chan, Kaifang Zhou.





- Lecture: Zoltan Szabo



Introduction: Phil, Kaifang

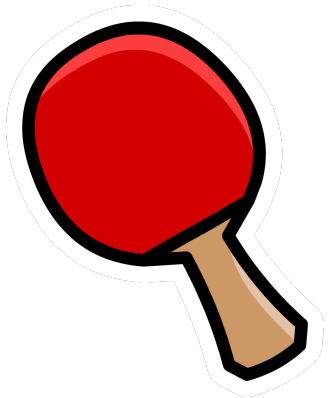
1. Phil (3 groups):
   - guest lecturer,
   - class teaching: Programming for Data Science (ST101), Probability and Distribution Theory (ST206), Statistical Inference (ST216), Regression and Generalised Linear Models (ST300), Machine Learning (ST310), Artificial Intelligence (ST311), ML and Data Mining (ST443), RL (ST455).
2. Kaifang (1 group):
   - LSE Fellow,
   - class teaching: Elementary Statistical Theory (ST102), Quantitative Methods (Statistics; ST107), Sample Surveys and Experiments (ST205), Stochastic Processes (ST302), Stochastic Simulation - computer workshop (ST303), ML and Data Mining (ST443), Managing and Visualising Data (ST445), Distributed Computing for Big Data (ST446).
   - research interests: ranking algorithms, choice models.

Introduction: Zoltan

1. background: applied maths, a bit of CS,
2. research interests: stat.ML $\rightarrow$ information theoretical estimators, kernel techniques, scalable computation & their applications,
3. hobby: nature; canoeing, photography, ping-pong.



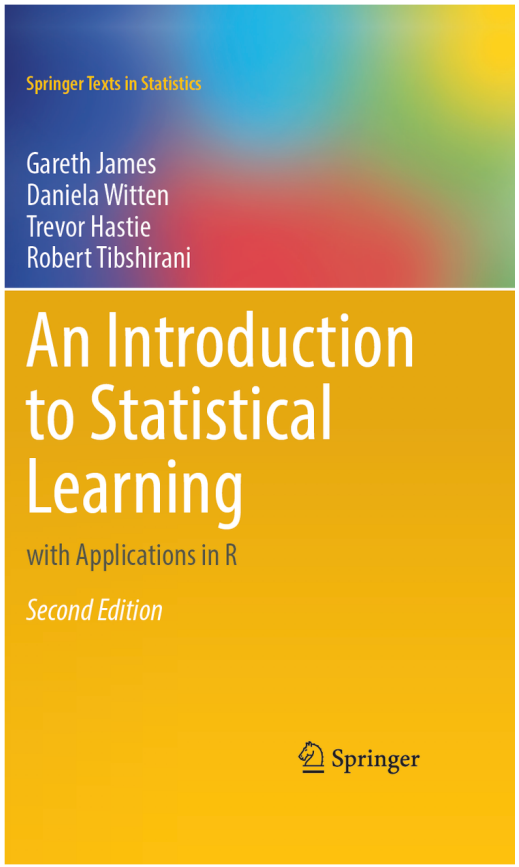


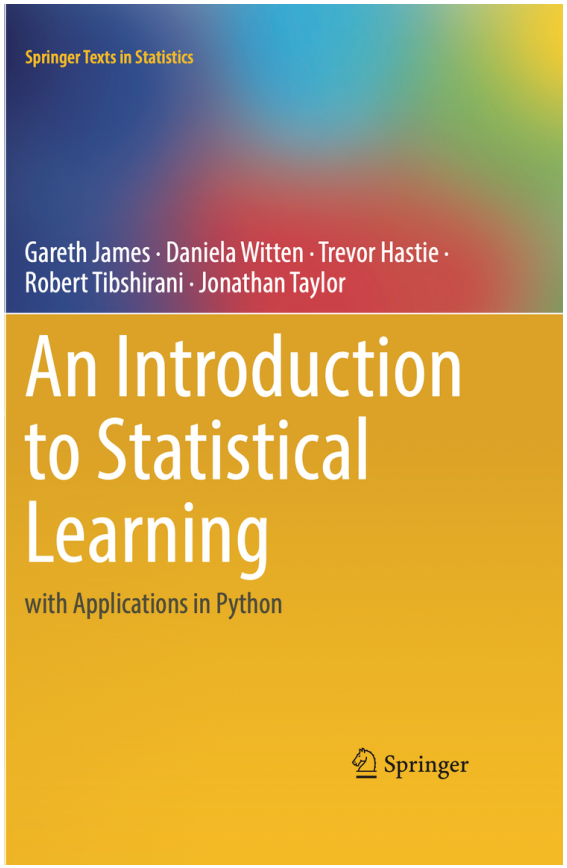


References & authors

Ref-1: backbone; minor branch outs when need arises

Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Jonathan Taylor (Python). An Introduction to Statistical Learning with Applications in R/Python, Springer, 2021 (R) and 2023 (Python) =: ISLR/ISLP.





A few words about the authors

Gareth James:



- Dean of UCLA Anderson School of Management.
- research:
  - statistics, data science, functional data analysis, marketing.
  - Google Scholar: $47,607$ citations.
- his PhD advisor (1998): Trevor Hastie.

Authors continued

Daniela Witten:



- Professor of Statistics & of Biostatistics @ University of Washington.
- research:
  - statistics, machine learning; highD statistics, selective inference, multiview data; genomics, neuroscience, microbial ecology, pathology.
  - Google Scholar: $68,873$ citations.

Authors continued

Trevor Hastie:



- Professor of Statistics & of Biomedical Data Science @ Stanford University.
- research:
  - statistical learning and modelling, data mining, machine learning.
  - Google Scholar: $468,975$ citations.

Authors continued

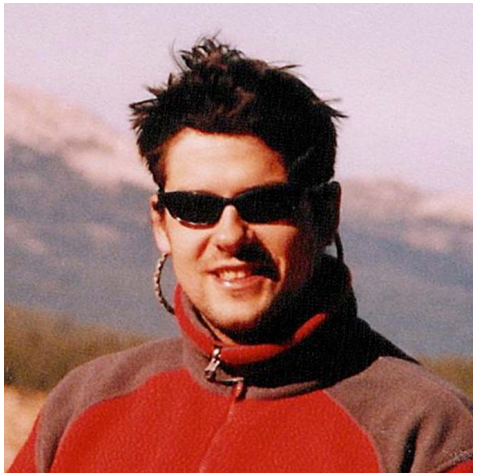
Robert Tibshirani:



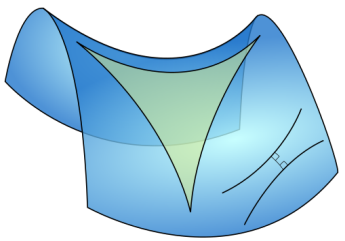
- Professor of Biomedical Data Science & of Statistics @ Stanford University.
- research:
  - statistics, data science, maching learning
  - Google Scholar: 633,635 citations.

Authors continued

Jonathan Taylor:

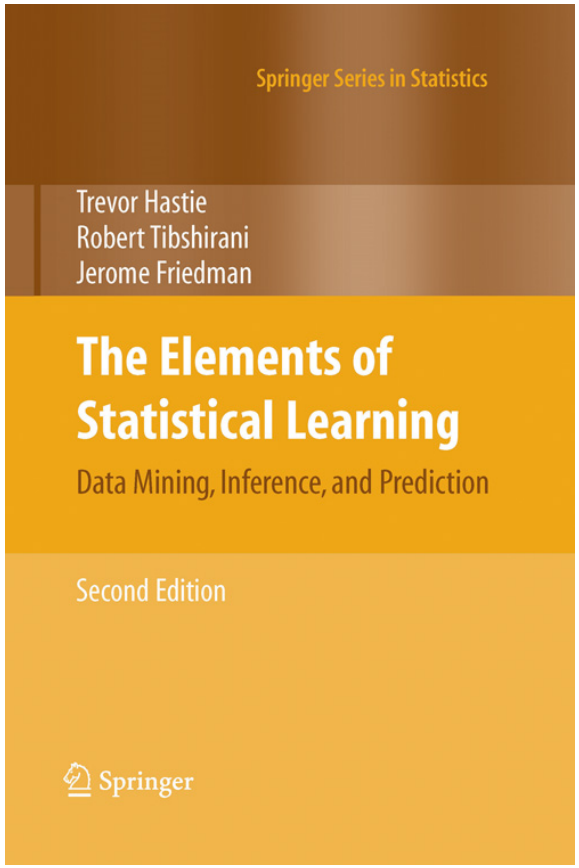


- Professor of Statistics @ Stanford University.
- research:
  - Gaussian processes, stochastic processes, differential geometry; neuroimaging, HIV protein sequence analysis.
  - Google Scholar: 75,781 citations.



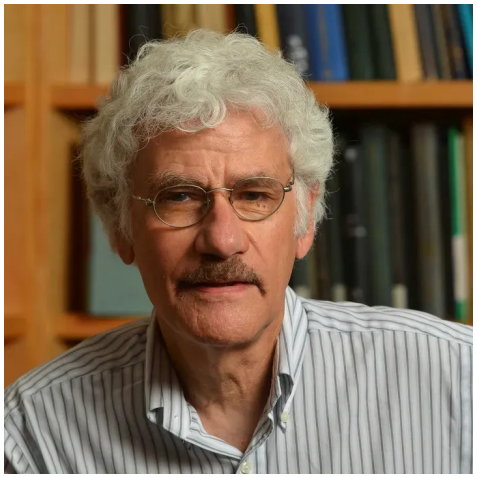
Ref-2: ESL

Trevor Hastie, Robert Tibshirani, Jerome Friedman. Elements of Statistical Learning, Springer, 2009.



Authors continued

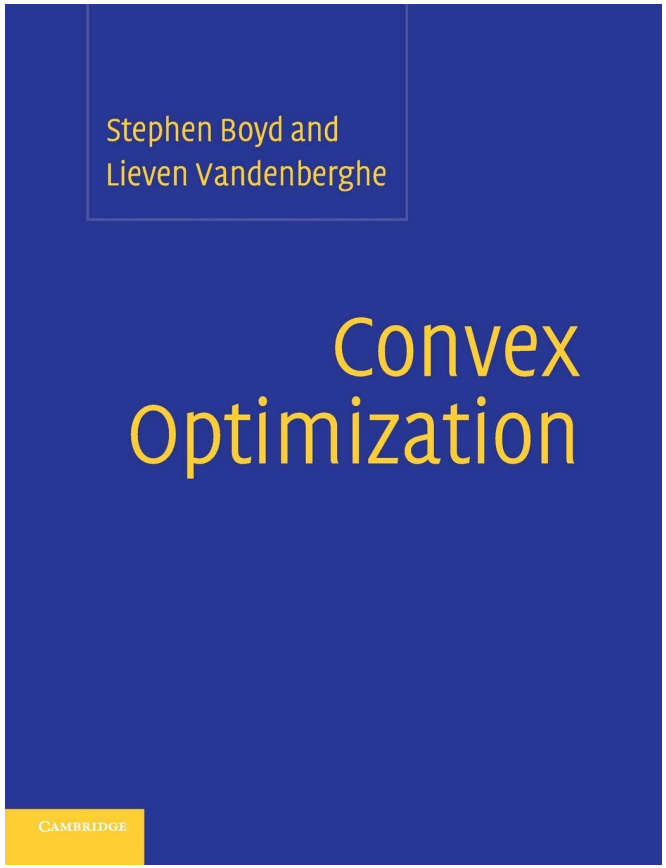
Jerome H. Friedman:



- Professor Emeritus of Statistics @ Stanford University.
- research:
  - statistics, data mining.
  - Google Scholar: 395,023 citations.

Ref-3: optimization, duality

Stephen Boyd and Lieven Vandenberghe. Convex Optimization, Cambridge University Press, 2004.



Authors continued

Stephen Boyd:



- Professor of Electrical Engineering, Computer Science, and Management Science @ Stanford University.
- research:
  - optimization, control, signal processing, AI, finance.
  - Google Scholar: $300,921$ citations.

Authors continued

Lieven Vandenberghe:



- Professor @ Electrical and Computer Engineering Department & Department of Mathematics, University of California, Los Angeles.
- research:
  - optimization, systems and control, signal processing.
  - Google Scholar: $115,968$ citations.

Ref-4: non-smooth optimization

- Neal Parikh and Stephen Boyd. Proximal Algorithms. Foundations and Trends in Optimization, volume 1, issue 3, pp. 127-239, 2014.
- Authors continued (Neal Parikh):



  - Adjunct Associate Professor @ School of International & Public Affairs, Columbia University.
  - research:
    - machine learning, convex optimization, statistics, technology policy, AI ethics.
    - Google Scholar: $37,515$ citations.

ML hints

Typesetting: LaTeX

- high-quality typesetting system (TeX) with macro packages (LaTeX),
- creator:
  - The Art of Computer Programming (book, 7 volumes): publisher with poor typesetting $\Rightarrow$ Donald Knuth (TeX, '78),
  - version number: 3.141592653 (in 2021) $\rightarrow \pi$,
  - coding errors in TeX: \$2.56 = 256 cents, $\times 2$.



  - Leslie Lamport (LaTeX, '84).

TeX = high-quality article/book/TR/poster/...with minimal effort

LaTeX – continued

- gold standard in ML,
- cross-platform: Windows, MacOS, Linux,
- CTAN = packages, ...
- collaborative editing: Overleaf ($\leftrightarrow$ Git),
- style: WYSI$\underbrace{\text{WYW}}_{\text{what you want}}$ instead of (all)WYSI$\underbrace{\text{WYG}}_{\text{what you get}}$,
- 1/2h intro to LaTeX.

Top ML venues $\Rightarrow$ pick up the writing 'etiquette'

- Terminology:

  AI $\approx$ ML $\approx$ data science $\approx$ applied maths $\approx$ data-centric engineering
- Mathwise:

$$\underbrace{\text{statistics}}_{\text{guarantees}}, \underbrace{\text{optimization}}_{\text{computation}}, \underbrace{\text{functional analysis}}_{\text{function spaces}}$$

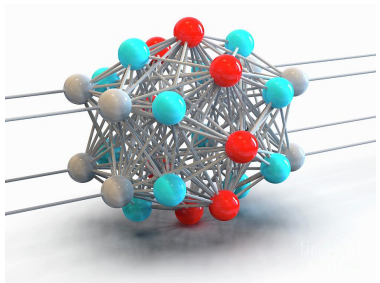
- Top-tier conferences [acceptance $\sim 20 - 25\%$]:
  - Conference on Neural Information Processing Systems (NeurIPS),
  - International Conference on Machine Learning (ICML),
  - International Conference on Artificial Intelligence and Statistics (AISTATS),
  - Conference on Uncertainty in Artificial Intelligence (UAI),
  - Conference on Learning Theory (COLT).
- Top journal :
  - Journal of Machine Learning Research (JMLR).

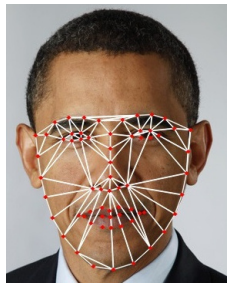
Top conferences+

Great 'app-specific' conferences:

- NLP:
  - Empirical Methods in Natural Language Processing (EMNLP).
- deep & representation learning:
  - International Conference on Learning Representations (ICLR).
- computer vision:
  - Conference on Computer Vision and Pattern Recognition (CVPR),
  - International Conference on Computer Vision (ICCV).







Publication morale, mentality

- conference-oriented! (vs stats),
- extension to journal (+theory/app),
- open (typically),
- reproducibility: more & more emphasis.

Preprint repository: arXiv

History:

- open-access repo of preprints,
- hosted by Cornell University,
- launched in 1991 (Paul Ginsparg; TEX made it possible),
- covered areas: mathematics, computer science, physics, quantitative biology, quantitative finance, statistics, electrical engineering and systems science, economics.
- submission rate (2025): 284,486 articles/year; further stats.

arXiv – continued

- Categories [1st approximation]:
  - stat.ML: statistical machine learning (stat),
  - cs.LG: learning (CS),
  - math.OC: optimization (maths).
- Description of sub-categories in statistics and CS, classification:
  - statistics, CS,
  - ML classification guide.
- Each paper has 1 primary and typically 1-2 secondary categories.
- In addition: MSC (=maths) and ACM (=CS) classes.

Mailing lists

- ML:
  - ML-News,
  - UAI.
- Statistics: allstat.
- ML + neuroscience: Connectionists.
- Academic social networking: LinkedIn.
- Computing job announcements: CRA.
- Jobs in the UK: jobs.ac.uk.

Programming languages

- Python:
  - dominant in data science.
  - general-purpose language $\xrightarrow{\text{specialized}}$ ML.
  - Good book: Mark Lutz. Learning Python, 6th Edition. O'Reilly Media, Inc., 2025.
- R:
  - widely-used in statistics.

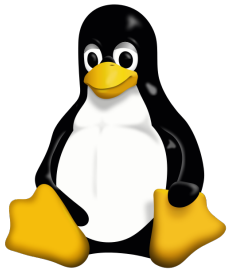
@ LSE

Digital Skills Lab: pre-sessional courses (Python, R, LaTeX).

OpSystem

- ? : Windows, MacOS, Linux, else.





- Linux:
  - idea: user freedom ⇒ you can own your computer.
  - free ⇒
    - versatile, sustainable, secure, privacy-respecting,
    - runs 'everywhere': laptops/desktops, mobiles, paper-like tablets, research computing (cluster), ...
- For the curious-minded:
  - Getting Started with Linux mini-course: slides (Sept. 25).

Git





Linus Torvalds: also author of Git,

- distributed version control system,
- standard in collaborative coding efforts,
- developed for the Linux kernel ('05),
- # of lines in the code of Linux kernel: 40+ million (2025).

OpenReview

- created by Andrew McCallum's lab (Univ. of Massachusetts Amherst),
- aim: open scientific communication, discussion,
- widely-used for ML conferences,
- roles: author $\rightarrow$ reviewer $\rightarrow$ AC $\rightarrow$ SAC $\rightarrow$ PC,
- process: submission, bidding, review, rebuttal & discussion, decision.



Git: continued; hosts: GitHub, GitLab, Bitbucket, ...

Relevant commands:

command	meaning
git clone url_of_repo	copy: remote → local (1×), xor
git pull	copy: remote → local (grab latest version)
git add .	add all modified/new files in current directory recursively
git commit -m "comment"	save changes (locally), create checkpoint
git push	copy: local → remote

Notes

- It can be integrated with Overleaf.
- They also allow webpage hosting.

Admin

Schedule, recording, office hours, email recipe

- Schedule:
  - W1 - 5, W7 - 11: lecture + seminar; W6: reading week.
- Recording:
  - No (unless you have adjustment in place & you show up on my LfY).
  - Take handwritten notes, it helps learning (not passive, no multi-tasking)!
- Office hours:
  - seminars: arrange with Phil and Kaifang.
  - lecture: (i) on-the-fly (here), (ii) contact by email ($\Rightarrow$ COL 5.14).
- Emails: send ST443-related emails to me with subject: [ST443] your topic !

Assessment

Exam (100%; 120 minutes in the Spring exam period; term dates).

- understanding of the learned concepts, computing, proving,
- hint: put a paper with a pen & puzzle next to the bed.
- we use/apply: linear algebra, real analysis, probability.
- basis of marking: detailed reasoning; a systematic way

$$A \overset{(a)}{=} B \overset{(b)}{\leq} C \ldots$$

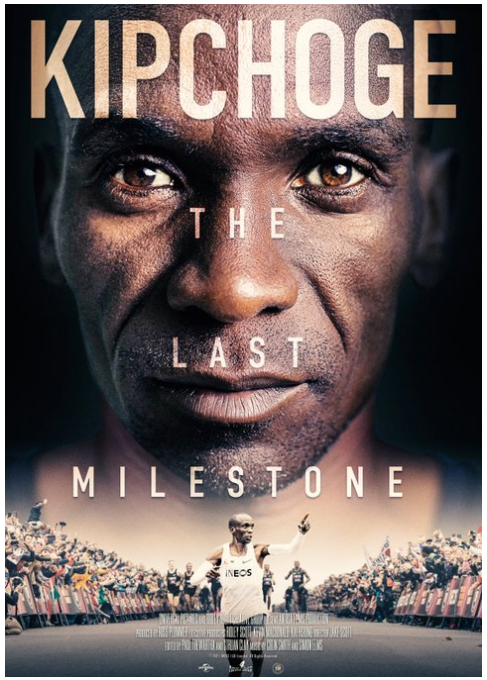
where A follows from X, B holds by Y.

Admin: genAI policy

GenAI policy

- We expect: what you submit is your work.
- Aim:
  - to demonstrate the knowledge and skills you have learned,
  - to obtain feedback for improvement.
- Course policy: no GenAI! (treated as academic misconduct)

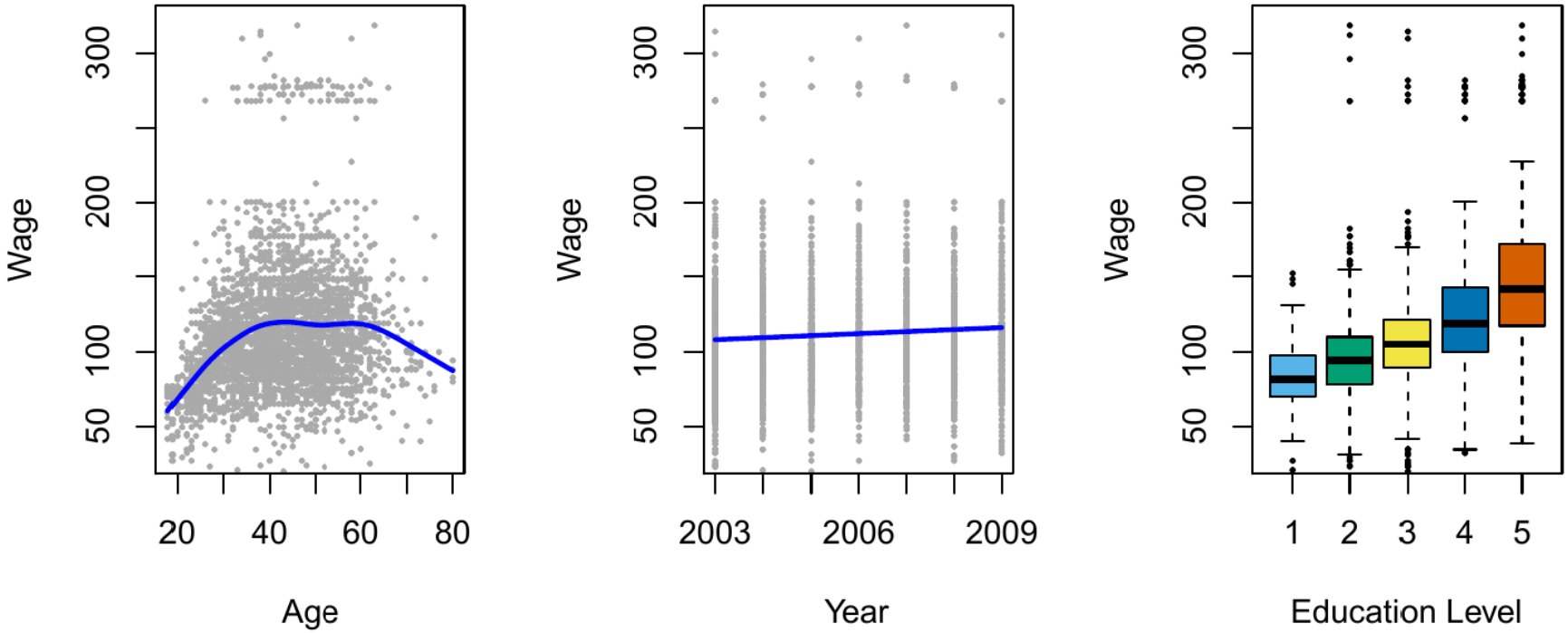




High-level idea (@ ISLP: Ch1)

Prototype examples: task-1

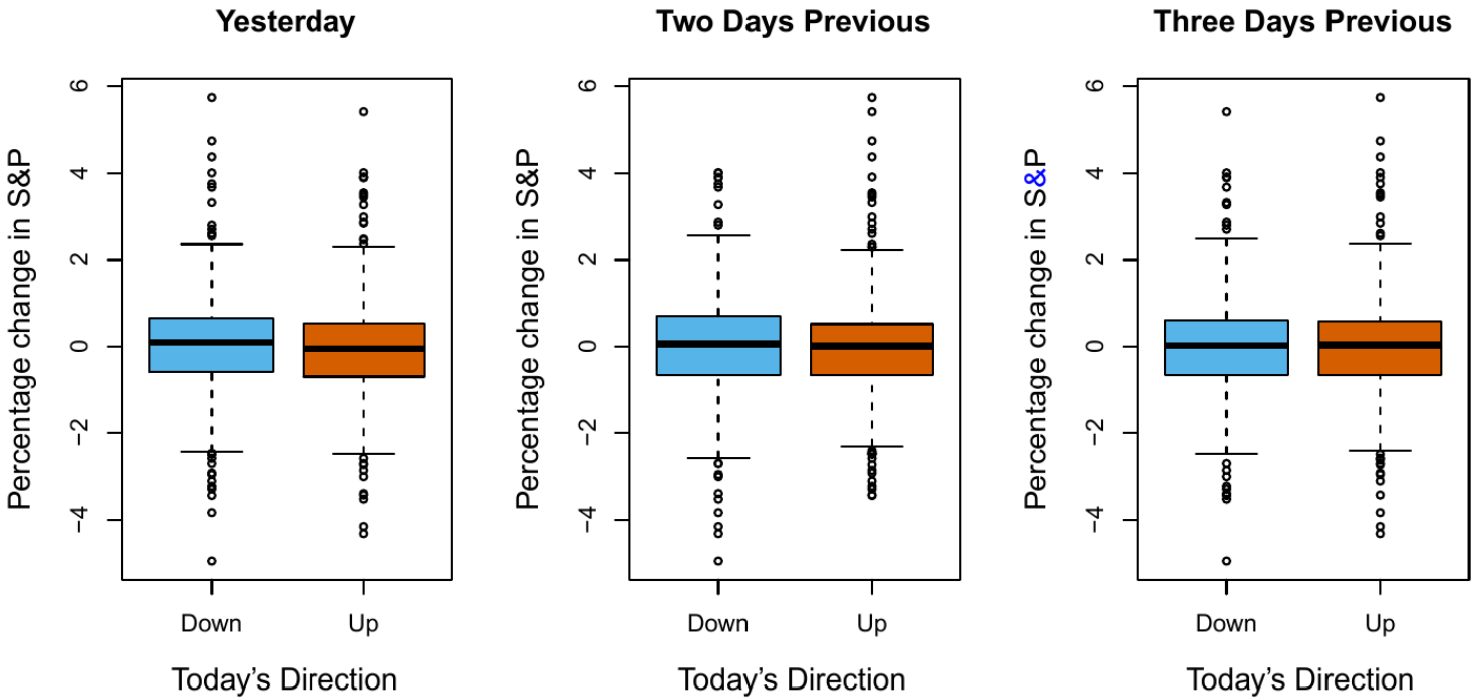
Aim: understand wage vs age, calendar year, education level association.



- Wage increases with
  - age, but decreases from ~ 60. $\Rightarrow$ fit a curve to predict? significant variation $\Rightarrow$ other variables can come handy.
  - year: increases roughly linearly (+\$10K), grows with education level.
- Goal: capture linear & non-linear relations; regression.

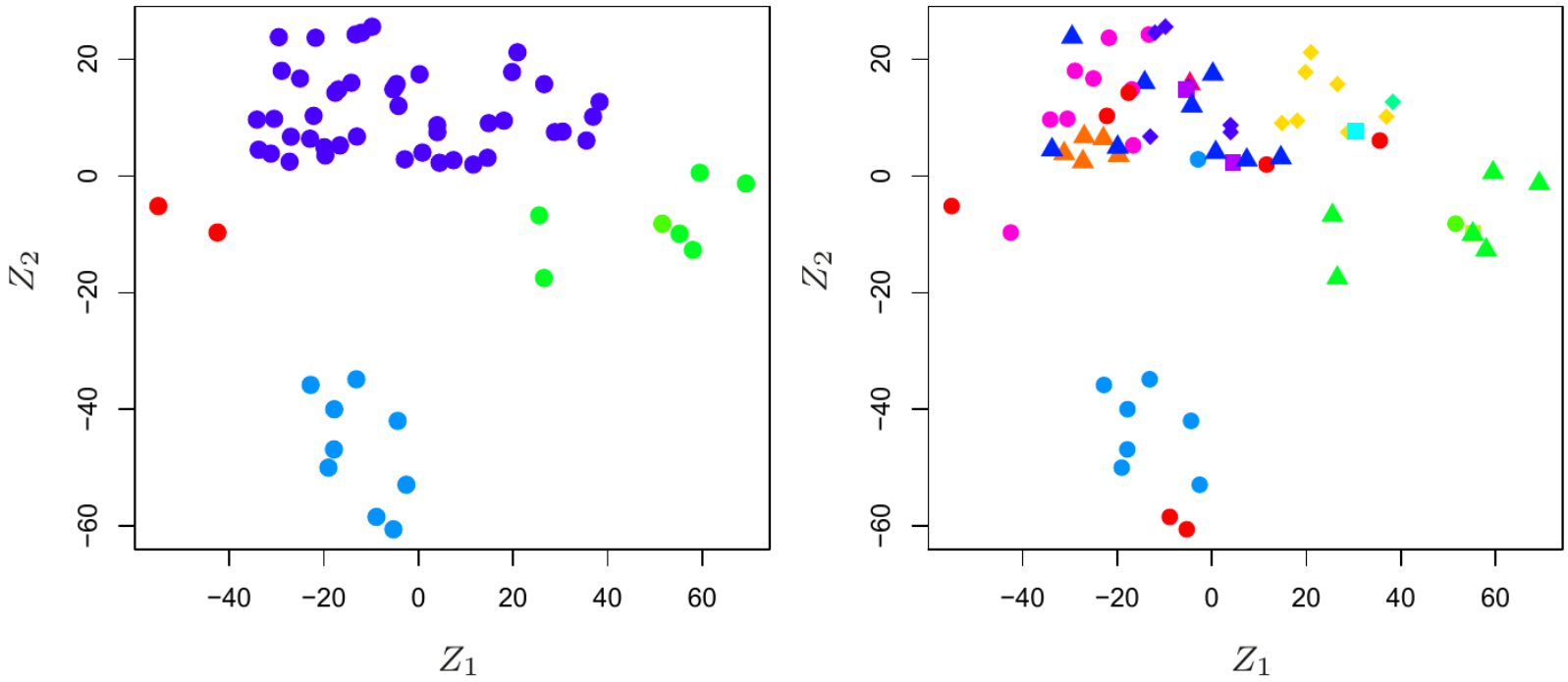
ISL: task-2

- Previous task: continuous output. Now: categorical/qualitative output.
- Aim: predict if S&P 500 will go up/down based on historical % change.
- Hard task, but pays well.;) classification (ex+: spam filtering).



ISL: task-3

- Previous examples: input-output pairs. Now: output is missing.
- Aim:
  - marketing: categorize customers by their observed characteristics.
  - biology: gene expression categorization ($n = 6830$, $p = 40$).
  - dimensionality reduction (visualization, PCA), clustering.



(Linear) algebra: recap

Typical representation: vector, vectorspace (over $\mathbb{R}$)

$(\mathbb{R}^d, +, \cdot)$:

$$(x_1, \ldots, x_d) + (y_1, \ldots, y_d) := (x_1 + y_1, \ldots, x_d + y_d),$$
$$\lambda \cdot (x_1, \ldots, x_d) := (\lambda x_1, \ldots, \lambda x_d).$$

2 operations on V: $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \quad \text{(associativity)}$$
$$v_1 + v_2 = v_2 + v_1, \quad \text{(commutativity)}$$
$$\exists 0 : v + 0 = v,$$
$$\exists - v : v + (-v) = 0,$$
$$a \cdot (b \cdot v) = (ab) \cdot v,$$
$$1 \cdot v = v,$$
$$a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2,$$
$$(a + b) \cdot v = a \cdot v + b \cdot v \quad \text{for } \forall v_1, v_2, v_3, v \in V, a, b \in \mathbb{R}.$$

Vectorspace: other examples

1. Polynomials $= \mathbb{R}[x]$ (finite sums):

$$\sum_i a_i x^i + \sum_i b_i x^i := \sum_i (a_i + b_i) x^i,$$

$$\lambda \cdot \left(\sum_i a_i x^i \right) := \sum_i (\lambda a_i) x^i.$$

2. $\mathcal{X} \to \mathbb{R}$-valued functions:

$$(f + g)(x) := f(x) + g(x), \qquad (\lambda \cdot f)(x) := \lambda f(x).$$

Note

1st example [with $\leq (d-1)$-order polynomials]:
$\mathcal{X} = [d] := \{1, \ldots, d\}$.

Vectors+

3. $\supset C[a,b]$ ($\mathcal{X} = [a,b]$): continuous functions.
4. $(\mathbb{R}^d, \otimes, \cdot)$: NO!!!

$$(x_1, \ldots, x_d) \otimes (y_1, \ldots, y_d) := [x_i y_j]_{i,j=1}^d \in \mathbb{R}^{d \times d} \neq \mathbb{R}^d,$$
$$\lambda \cdot (x_1, \ldots, x_d) := (\lambda x_1, \ldots, \lambda x_d).$$

5. $(\mathbb{R}^d, \langle \cdot, \cdot \rangle, \cdot)$: similarly NO

$$\langle (x_1, \ldots, x_d) \, , (y_1, \ldots, y_d) \rangle := \sum_{i=1}^d x_i y_i \in \mathbb{R} \neq \mathbb{R}^d,$$
$$\lambda \cdot (x_1, \ldots, x_d) := (\lambda x_1, \ldots, \lambda x_d).$$

Similarity: inner product

What is the similarity of 2 vectors?

A natural candidate: $\langle \mathbf{a}, \mathbf{b} \rangle := \sum_{i \in [d]} a_i b_i$ $(\mathbf{a}, \mathbf{b} \in \mathbb{R}^d)$

$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ is called **inner product** if for $\forall \lambda_i \in \mathbb{R}, v_i, v, w \in V$

1. $\langle \lambda_1 v_1 + \lambda_2 v_2, w \rangle = \lambda_1 \langle v_1, w \rangle + \lambda_2 \langle v_2, w \rangle$ (linearity),
2. $\langle v, w \rangle = \langle w, v \rangle$ (symmetry),
3. $\langle v, v \rangle \geq 0$; $\langle v, v \rangle = 0 \Leftrightarrow v = 0$. (positive definiteness).

Similarity: inner product $\Rightarrow$ Euclidean space

- $(V, \langle \cdot, \cdot \rangle)$ is called **Euclidean space**.
- Examples beyond $\mathbb{R}^d$:
  - $\left(\mathbb{R}^{d_1 \times d_2}, \langle \mathbf{A}, \mathbf{B} \rangle_{\mathsf{F}} = \mathrm{tr}(\mathbf{A}^\top \mathbf{B}) = \sum_{i \in [d_1], j \in [d_2]} A_{i,j} B_{i,j}\right)$.
  - $\left(C[a, b], \langle f, g \rangle = \int_a^b f(x) g(x) \mathrm{d}x\right)$.

Similarity: norm, angle

Why do we like inner products?

It induces a notion of

- length (norm): $\|v\| := \sqrt{\langle v, v \rangle}$.
- angle : CBS inequality

$$\langle v, w \rangle \leq \|v\| \, \|w\| \, .$$

Hence, for $v, w \neq 0$,

$$[-1, 1] \ni \frac{\langle v, w \rangle}{\|v\| \, \|w\|} =: \cos(v, w).$$

Similarity: norm

What are the properties of (the induced / a) norm?

$\|\cdot\| : V \to [0, \infty)$ is called **norm** if

1. $\|v\| = 0$ iff. $v = 0$ (norm separates points),
2. $\|\lambda v\| = |\lambda| \, \|v\| \; \forall \lambda \in \mathbb{R}, \forall v \in V$ (positive homogenity),
3. $\|v + w\| \leq \|v\| + \|w\| \; \forall v, w \in V$ (triangle inequality).

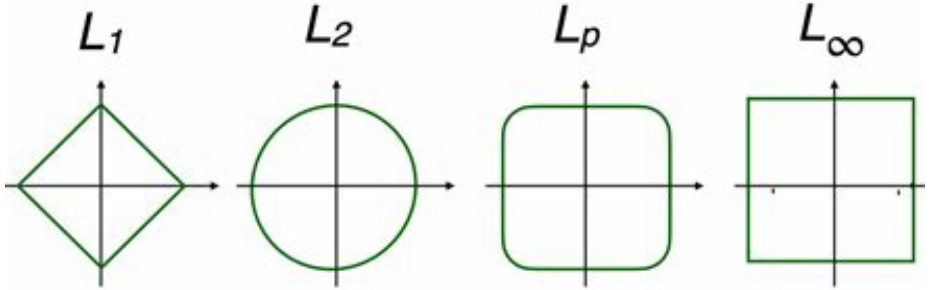
Note:

- norm $\Rightarrow$ metric: $\rho(f, g) = \|f - g\| \Rightarrow$
- allows one to study continuity, convergence.

Similarity: norm $\Rightarrow$ normed space

$(V, \|\cdot\|)$ is called **normed space**. Examples:

- $(\mathbb{R}, |\cdot|)$.
- $\left(\mathbb{R}^d, \|\mathbf{x}\|_p = \left[\sum_{i\in[d]} |x_i|^p\right]^{\frac{1}{p}}\right)$, $p \in [1, \infty)$:
  - $p = 2$: $\|\mathbf{x}\|_2 = \sqrt{\sum_{i\in[d]} x_i^2}$ (Euclidean),
  - $p = 1$: $\|\mathbf{x}\|_1 = \sum_{i\in[d]} |x_i|$ (Manhattan),
  - $p = \infty$: $\|\mathbf{x}\|_\infty := \lim_{p\to\infty} \|\mathbf{x}\|_p = \max_{i\in[d]} |x_i|$ (maximum norm).

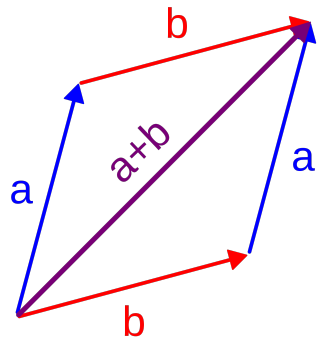


- $\left(C[a, b], \|f\|_p = \left[\int_a^b |f(x)|^p \mathrm{d}x\right]^{\frac{1}{p}}\right)$, $1 \le p < \infty$.
- $\left(C[a, b], \|f\|_\infty = \max_{x\in[a,b]} |f(x)|\right)$, $p = \infty$.

Similarity: norm vs inner product

Relations:

- $|\langle a, b\rangle| \leq \|a\|\|b\|$ (CBS),
- $4\langle a, b\rangle = \|a+b\|^2 - \|a-b\|^2$ (polarization identity),
- $\|a+b\|^2 + \|a-b\|^2 = (*)$ (parallelogram rule).



$$(*) = \langle a+b, a+b\rangle + \langle a-b, a-b\rangle$$
$$= 2\left(\|a\|^2 + \|b\|^2\right) \pm 2\langle a, b\rangle = 2\|a\|^2 + 2\|b\|^2.$$

The parallelogram rule holds in an inner product space.

Parallelogram rule: 'proof template'

Let V be a Euclidean space. Show that the parallelogram rule holds in V:

$$\|f+g\|^2 + \|f-g\|^2 = 2\left[\|f\|^2 + \|g\|^2\right] \quad (f,g \in V).$$

Proof: In a Euclidean space the norm is induced by the inner product [i.e., $\|v\| = \sqrt{\langle v, v\rangle}$ for all $v \in V$] which means that

$$\|v\|^2 = \langle v, v\rangle \quad (v \in V). \tag{1}$$

Then, for all $f, g \in V$, one has

$$\begin{aligned}
\|f+g\|^2 + \|f-g\|^2 &\overset{(a)}{=} \langle f+g, f+g\rangle + \langle f-g, f-g\rangle \\
&\overset{(b)}{=} \langle f,f\rangle + \langle f,g\rangle + \langle g,f\rangle + \langle g,g\rangle + \\
&\qquad \langle f,f\rangle - \langle f,g\rangle - \langle g,f\rangle + \langle g,g\rangle \\
&\overset{(c)}{=} 2\left[\langle f,f\rangle + \langle g,g\rangle\right] \\
&\overset{(d)}{=} 2\left[\|f\|^2 + \|g\|^2\right],
\end{aligned}$$

where (a) follows from (1), (b) is by the linearity of the inner product, the terms were collected in (c), in (d) we used (1) again. This shows the claim.

Example when the parallelogram rule fails

$C[0,1]$ with $\|f\|_\infty = \max_{x\in[0,1]} |f(x)|$:

$$f(x) := 1 - x,\ g(x) := x,$$

$$\|f+g\|_\infty^2 + \|f-g\|_\infty^2 = \|1\|_\infty^2 + \|\underbrace{1-2x}_{\in[-1,1]}\|_\infty^2 = 1^2 + 1^2 = 2,$$

$$2\left(\|f\|_\infty^2 + \|g\|_\infty^2\right) = 2\|\underbrace{1-x}_{\in[0,1]}\|_\infty^2 + 2\|\underbrace{x}_{\in[0,1]}\|_\infty^2 = 2+2 = 4.$$

Characterization

A norm is induced by an inner product iff

$$\|f+g\|^2 + \|f-g\|^2 = 2\left(\|f\|^2 + \|g\|^2\right) \quad \forall f, g.$$

- Motivation:

$$x_1 + 2x_2 = 5,$$
$$3x_1 + 4x_2 = 6.$$

- One is lazy to write this much. $\Rightarrow$ With matrices:

$$\mathbf{Ax} = \mathbf{b}, \qquad \mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Data: basic matrix operations

- $\mathbf{A} + \mathbf{B}$, $\lambda\mathbf{A}$ (vector space): $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{d_1 \times d_2}$, $\lambda \in \mathbb{R}$,

$$\mathbf{A} + \mathbf{B} := [A_{i,j} + B_{i,j}]_{i,j=1,1}^{d_1,d_2} \in \mathbb{R}^{d_1 \times d_2}, \quad \lambda\mathbf{A} := [\lambda A_{i,j}]_{i,j=1}^{d_1,d_2} \in \mathbb{R}^{d_1 \times d_2}.$$

- $\mathbf{AB}$ (ring): $\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$, $\mathbf{B} \in \mathbb{R}^{d_2 \times d_3}$, defined with inner product

$$\mathbf{AB} = [\langle A_{i,:} B_{:,j} \rangle]_{i,j=1,1}^{d_1,d_3} \in \mathbb{R}^{d_1 \times d_3}.$$

Data: matrix – continued

- $\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$ is an $\mathbb{R}^{d_2} \to \mathbb{R}^{d_1}$ linear map

$$\mathbf{A}(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{A}\mathbf{v}_1 + \mathbf{A}\mathbf{v}_2, \qquad \mathbf{A}(\lambda\mathbf{v}) = \lambda(\mathbf{A}\mathbf{v}).$$

- Functional analysis: One works similarly on vector/Euclidean/normed/Hilbert/Banach spaces.

Pseudo-inverse: $\exists!$, characterization

- For any $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a unique $\mathbf{A}^- \in \mathbb{R}^{n \times m}$:

$$\begin{aligned} \mathbf{A}\mathbf{A}^-\mathbf{A} &= \mathbf{A}, \\ \mathbf{A}^-\mathbf{A}\mathbf{A}^- &= \mathbf{A}^-, \\ (\mathbf{A}\mathbf{A}^-)^\top &= \mathbf{A}\mathbf{A}^-, \\ (\mathbf{A}^-\mathbf{A})^\top &= (\mathbf{A}^-\mathbf{A}). \end{aligned}$$

Remains

Why is $\mathbf{A}^-$ useful? How to compute it?

Pseudo-inverse: usefulness – an example

Problem: solution of

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

might not exist or may not be unique.

The pseudo-inverse based solution $\hat{\mathbf{x}} = \mathbf{A}^{-}\mathbf{b}$ handles this issue and it is 'optimal'.

- smallest error: for any $\mathbf{x} \in \mathbb{R}^n$

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2 \geq \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|_2.$$

- smallest norm: among the $\mathbf{x}$ vectors for which '$=$' holds $\hat{\mathbf{x}}$ has minimal Euclidean norm.

Pseudo-inverse: properties

- Generalization of inverse: if $\mathbf{A}^{-1}$ exists, then $\mathbf{A}^{-} = \mathbf{A}^{-1}$.
- For $\mathbb{R}^{m\times n} \ni \mathbf{0}$: $\mathbf{0}^{-} = \mathbf{0} \in \mathbb{R}^{n\times m}$.
- For $\mathbf{\Sigma} = \mathrm{diag}(\sigma_i) := \mathrm{diag}\left((\sigma_i)_{i=1}^{\min(m,n)}\right) \in \mathbb{R}^{m\times n}$ diagonal:

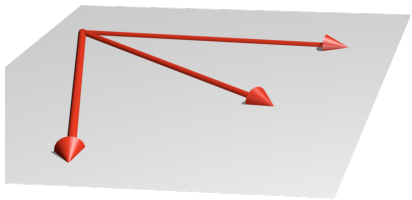
$$\mathbf{\Sigma}^{-} = \mathrm{diag}(\sigma_i^{-}) \in \mathbb{R}^{n\times m},$$

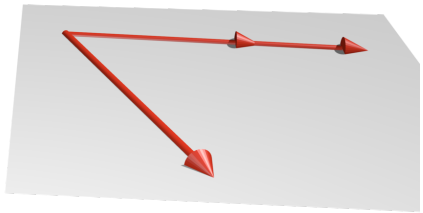
$$\sigma^{-} = \begin{cases} \frac{1}{\sigma} & \text{if } \sigma \neq 0, \\ 0 & \text{if } \sigma = 0. \end{cases}$$

- $(\mathbf{A}^{-})^{-} = \mathbf{A}$.
- It commutes with transposition: $(\mathbf{A}^{\top})^{-} = (\mathbf{A}^{-})^{\top}$.
- With scalar multiplication: $(c\mathbf{A})^{-} = \frac{1}{c}\mathbf{A}^{-}$ with $c \neq 0$.

Pseudo-inverse: in case of linear independence

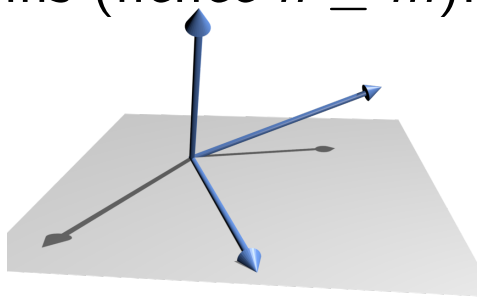
Linearly dependent vectors (2 examples):





- If $\mathbf{A} \in \mathbb{R}^{m \times n}$ has linearly independent columns (hence $n \leq m$):

$$\mathbf{A}^{-} = \left(\mathbf{A}^{\top}\mathbf{A}\right)^{-1}\mathbf{A}^{\top}.$$



PCA: ?

- If $\mathbf{A}$ has linearly independent rows:

$$\mathbf{A}^{-} = \mathbf{A}^{\top}\left(\mathbf{A}\mathbf{A}^{\top}\right)^{-1}.$$

- Gaussian elimination (GE) $\Rightarrow$ checking linear (in)dependence, rank, solving linear equation, matrix inversion ($\mathbf{A}|\mathbf{I} \xrightarrow{\text{GE}} \mathbf{I}|\mathbf{A}^{-1}$).

Pseudo-inverse: computation

One can get $\mathbf{A}^-$ from the SVD of $\mathbf{A}$.

For an $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

factorization (rotate $\circ$ scale $\circ$ rotate) , where

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ is orthogonal: $\mathbf{U}^\top\mathbf{U} = \mathbf{I}$. (intuition? general form? Givens rotation? optimization?)
- $\mathbf{V} \in \mathbb{R}^{n \times n}$ is orthogonal.
- $\mathbf{\Sigma} = \mathrm{diag}(\sigma_i) \in \mathbb{R}^{m \times n}$ is diagonal with non-negative entries, $\searrow$.
- $(\sigma_i)_{i=1}^{\min(m,n)}$: singular values.

SVD $\Rightarrow$ pseudoinverse

$\mathbf{A}^- = \mathbf{V}\mathbf{\Sigma}^-\mathbf{U}^\top$.

SVD reveals 'everything' about the matrix

Examples:

- $\text{rank}(\mathbf{A}) = \#$ of non-zero σ_i-s.
- $\|\mathbf{A}\|_F = \sqrt{\sum_i \sigma_i^2(\mathbf{A})} = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{tr}(\mathbf{A}^\top \mathbf{A})}$: Frobenius (a.k.a. Hilbert-Schmidt) norm. Note: $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.
- $\|\mathbf{A}\| = \sigma_1(\mathbf{A})$: operator norm.
- Schatten p-norm:

$$\|\mathbf{A}\|_p = \|(\sigma_i(\mathbf{A}))_i\|_p, \quad p \in [1, \infty].$$

Notes:

- Spec.: $p = 2$ (Frobenius), $p = \infty$ (operator norm), $p = 1$ (nuclear / trace norm; ?, Lasso: ?).
- $\|\mathbf{A}\|_\infty \leq \|\mathbf{A}\|_2 \leq \|\mathbf{A}\|_1$.

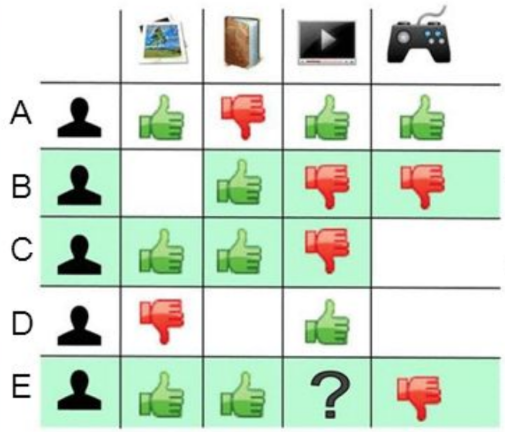
Nuclear norm: intuition

- Low-rank view:

$$\mathbf{A} = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^\top,$$

$$\|\mathbf{A}\|_1 = \|\boldsymbol{\sigma}\|_1 = \sum_i \underbrace{|\sigma_i|}_{\sigma_i}.$$

- $\|\boldsymbol{\sigma}\|_1$ 'captures' the # of non-zero σ_i-s. $\Rightarrow$
- Low-rank structures (collaborative filtering):



SVD note

Low-rank approximation problem [recall: $\|\mathbf{M}\|_{\mathrm{F}}^2 = \mathrm{tr}\left(\mathbf{M}^\top \mathbf{M}\right)$]

- Let $\mathbb{R}^{a\times b} \ni \mathbf{M}$.
- Task: with $k \leq r := \mathrm{rank}(\mathbf{M})$, $\min_{\mathbf{X}\in\mathbb{R}^{a\times b}\,:\,\mathrm{rank}(\mathbf{X})=k} \|\mathbf{X}-\mathbf{M}\|_{\mathrm{F}}^2$.
- Solution (best k-rank approximation):

$$\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top\text{: SVD of } \mathbf{M},$$
$$\mathbf{D} = \mathrm{diag}(\sigma_1,\ldots,\sigma_r,0,\ldots,0), \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0,$$
$$\mathbf{X}_k = \mathbf{U}\hat{\mathbf{D}}_k\mathbf{V}^\top, \quad \hat{\mathbf{D}}_k = \mathrm{diag}(\sigma_1,\ldots,\sigma_k,0,\ldots,0).$$

Solution of low-rank approximation

Proof idea

1. Orthogonal invariance of $\|\cdot\|_{\mathsf{F}}$: for any $\mathbf{A} \in \mathbb{R}^{c \times d}$

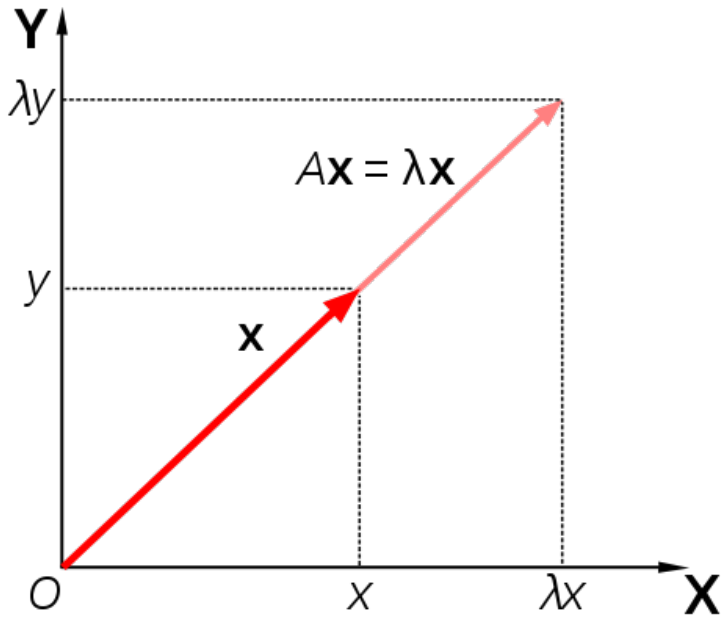
$$\left\|\mathbf{U}\mathbf{A}\mathbf{V}^{\top}\right\|_{\mathsf{F}} = \|\mathbf{A}\|_{\mathsf{F}}, \quad \forall \mathbf{U} \in O(c), \mathbf{V} \in O(d).$$

2. Similarly, rank is orthogonally invariant.
3. The problem can be reduced to the diagonal $\mathbf{D}$.

Eigensystems: eigenvectors, eigenvalues

- Simplest transformation: scaling.
- $\mathbf{x} \neq \mathbf{0}$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda \in \mathbb{R}$ if

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}.$$



- $\mathbf{A} = \mathbf{A}^\top \Rightarrow$ all $\lambda(\mathbf{A}) \in \mathbb{R}$.
- Size of $\mathbf{A}$? Eigenvalues: roots of $k_{\mathbf{A}}(\lambda) =$ det $(\mathbf{A} - \lambda\mathbf{I})$, ?.

Determinant

A few properties with $\mathbf{A} \in \mathbb{R}^{n \times n}$:

- Formal definition:

$$\det(\mathbf{A}) = \sum_{\sigma=(\sigma_i)_{i=1}^n \in S_n} \text{sign}(\sigma) a_{1,\sigma_1} a_{2,\sigma_2} \dots a_{n,\sigma_n},$$

  where $\text{sign}(\sigma) := (-1)^{N(\sigma)}$ is the signature of the permutation σ,

  $N(\sigma) = \#$ of inversions in $\sigma = \#\{i < j \text{ and } \sigma(i) > \sigma(j) \ : \ i, j \in [n]\}$.

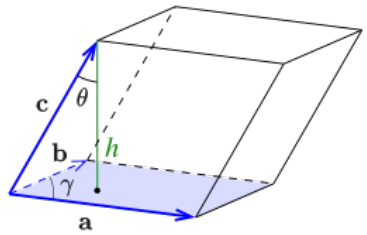
- For $n = 1, 2$: simple explicit formula,

$$\det([a]) = a, \qquad \det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc.$$

- Generally: recursion (Laplace expansion).

Determinant – continued

- Geometric intuition: (signed) volume of the parallelepiped spanned by the columns of $\mathbf{A}$ ($= [\mathbf{a}, \mathbf{b}, \mathbf{c}]$ below). Ex: rectangle (different orientations!).



- $\det(c\mathbf{A}) = c^n \det(\mathbf{A})$, $\det(\mathbf{A}) = \det\left(\mathbf{A}^\top\right)$, $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$.
- For $\mathbf{A}$ triangular: $\det(\mathbf{A}) = \prod_{i\in[n]} A_{i,i}$, $[n] := \{1, 2, \ldots, n\}$.
- Relation to eigenvalues: $\det(\mathbf{A}) = \prod_{i\in[n]} \lambda_i(\mathbf{A})$, $\lambda_i(\mathbf{A}) \in \mathbb{C}$.

Determinant computation: example; det($\mathbf{A}$)=?

Let $\mathbf{A}$ and our chessboard (which we visualize!) be

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \qquad \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Then, we get that

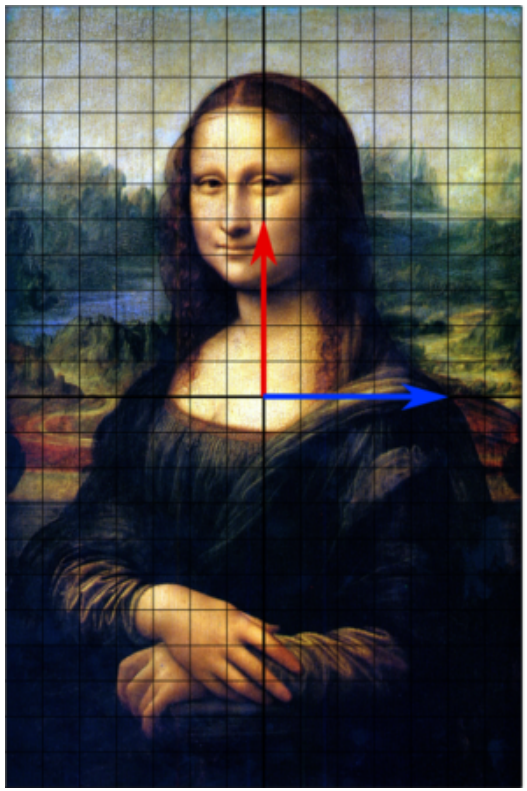
$$\begin{aligned} \det(\mathbf{A}) &\overset{(a)}{=} +1 \det\left(\begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}\right) -4 \det\left(\begin{bmatrix} 2 & 3 \\ 8 & 9 \end{bmatrix}\right) +7 \det\left(\begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix}\right) \\ &\overset{(b)}{=} (5 \cdot 9 - 6 \cdot 8) - 4(2 \cdot 9 - 3 \cdot 8) + 7(2 \cdot 6 - 3 \cdot 5) \\ &\overset{(c)}{=} (45 - 48) - 4(18 - 24) + 7(12 - 15) \overset{(c)}{=} -3 + 24 - 21 \overset{(c)}{=} 0, \end{aligned}$$

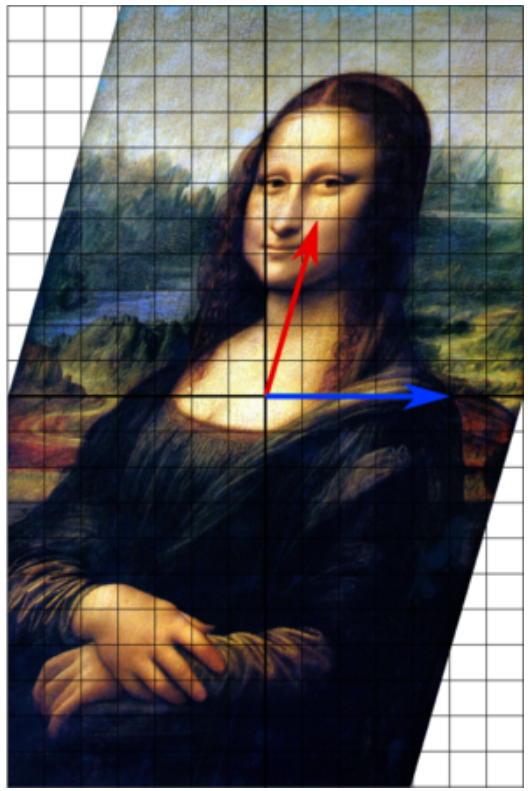
where in (a) we expanded (Laplace expansion) along the 1st column of $\mathbf{A}$, in (b) we used the explicit formula of determinants for 2×2 matrices, and simplified in (c). Notes: (i) det(2×2 matrix) is a specific case of the rule, (ii) alternative: GE.

Eigensystems: continued

Examples:

- Identity: $\mathbf{A} = \mathbf{I}$.
- Diagonal matrix: $\mathbf{A} = \text{diag}(a_i)$, spec.: reflection.
- Shear mapping on Mona Lisa:





Symmetric matrices are nice

- Diagonal matrix: the eigensystem can be chosen to be orthogonal – as we saw it.
- A symmetric $\mathbf{A}$ ($\mathbf{A} = \mathbf{A}^\top$) behaves similarly: $\exists$

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top,$$

where $\mathbf{\Sigma} = \text{diag}(\lambda_i)$, $\mathbf{U}$: orthogonal.

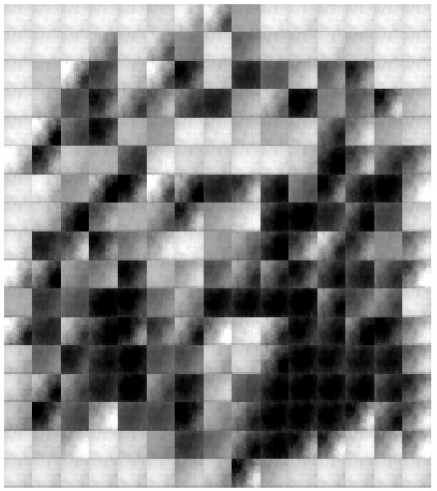
An application of eigensystems

- principal component analysis (PCA): demo now.

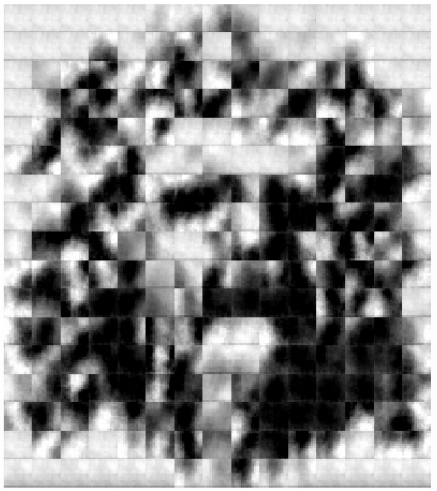
PCA example: 100% $\rightarrow$ 1%, 2%, 5%, 10%, 20%



(A)



(B)



(C)



(D)



(E)



(F)

Note: right vs. left eigenvector

Sizes: $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{x} \in \mathbb{R}^n$ (column vector)., $\mathbf{y} \in \mathbb{R}^{1 \times n}$ (row vector).

Eigenvector := right eigenvector

$\mathbf{A}\mathbf{x} = \lambda \mathbf{x}$.

Left eigenvector

$\mathbf{y}\mathbf{A} = \kappa \mathbf{y}$. $(\Leftrightarrow \mathbf{A}^\top \mathbf{y}^\top = \kappa \mathbf{y}^\top)$

In words:

- The left eigenvectors of $\mathbf{A}$ are the right eigenvectors of $\mathbf{A}^\top$ (modulo $\top$).
- The left and right eigenvalues of $\mathbf{A}$ are identical $\Leftarrow$

$$k_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I}_n) = \det\left((\mathbf{A} - \lambda \mathbf{I}_n)^\top\right) = \det\left(\mathbf{A}^\top - \lambda \mathbf{I}_n\right) = k_{\mathbf{A}^\top}(\lambda).$$

Local summary

Warmup:

- team, references & their authors, ML hints,
- admin,
- prototype tasks.
- linear algebra: recap.



Why does the Laplace expansion hold?

Warmup: Gaussian elimination for det($\mathbf{A}$) computation

Recipe:

1. apply elementary transformations to transform $\mathbf{A}$ to triangular,
2. compute the det of the resulting triangular matrix (product of diag),
3. update det in line with the applied elementary transformations.

Elementary transformations $[\mathbf{A} \to \tilde{\mathbf{A}}]$:

1. swapping 2 rows: $R_i \leftrightarrow R_j$,
2. multiplying a row with a $c(\neq 0)$ constant: $R_i \to cR_i$,
3. adding a scalar multiple of a row to an other row: $R_i \to R_i + c'R_j$.

det change for elementary transformations ($\tilde{\mathbf{A}} = \mathbf{EA}$)

transformation ($\mathbf{E}$)	det change
$R_i \leftrightarrow R_j$	$\det(\tilde{\mathbf{A}}) = -\det(\mathbf{A})$
$R_i \to cR_i$	$\det(\tilde{\mathbf{A}}) = c\det(\mathbf{A})$
$R_i \to R_i + CR_j$	$\det(\tilde{\mathbf{A}}) = \det(\mathbf{A})$

Idea on the why: $\det(\tilde{\mathbf{A}}) = \det(\mathbf{E})\det(\mathbf{A})$ and lift to $\mathbf{I}_n$

$$\mathbf{C} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \qquad \det(\mathbf{C}) = 0 \times 0 - 1 \times 1 = -1,$$

$$\mathbf{M} = \begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \qquad \det(\mathbf{M}) = c \times 1 - 0 \times 0 = c,$$

$$\mathbf{S} = \begin{bmatrix} 1 & c' \\ 0 & 1 \end{bmatrix} \Rightarrow \qquad \det(\mathbf{S}) = 1 \times 1 - 0 \times c' = 1.$$

Example: Gaussian elimination for det $:= |\cdot|$

$$\left|\begin{bmatrix} 2 & 4 & 6 \\ 2 & 4 & 4 \\ 3 & 3 & 7 \end{bmatrix}\right| \overset{(a)}{=} \left|\begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & -2 \\ 3 & 3 & 7 \end{bmatrix}\right| \overset{(b)}{=} -\left|\begin{bmatrix} 2 & 4 & 6 \\ 3 & 3 & 7 \\ 0 & 0 & -2 \end{bmatrix}\right| \overset{(c)}{=} -2\left|\begin{bmatrix} 1 & 2 & 3 \\ 3 & 3 & 7 \\ 0 & 0 & -2 \end{bmatrix}\right|$$

$$\overset{(d)}{=} -2\left|\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{bmatrix}\right| = (-2)1(-3)2 = -12,$$

(a): $R_2 \to R_2 - R_1$, (b) $R_2 \leftrightarrow R_3$, (c) $R_1 \to \frac{1}{2}R_1$, (d) $R_2 \to R_2 - 3R_1$.

Computational complexity ?

Laplace expansion: $\mathcal{O}(n!)$; Gaussian elimination: $\mathcal{O}(n^3)$ (scales better).

Laplace expansion

Let $\mathbf{A} = [a_{i,j}]_{i,j=1}^{n} \in \mathbb{R}^{n \times n}$ with $n \geq 2$. Then,

- expansion along the i-th row: for all $i \in [n]$

$$|\mathbf{A}| = \sum_{j \in [n]} a_{i,j}(-1)^{i+j}|\mathbf{A}_{i,j}|,$$

- expansion along the j-th column: for all $j \in [n]$

$$|\mathbf{A}| = \sum_{i \in [n]} a_{i,j}(-1)^{i+j}|\mathbf{A}_{i,j}|,$$

where $\mathbf{A}_{i,j} \in \mathbb{R}^{(n-1)\times(n-1)}$ denotes $\mathbf{A}$ without the i-th row and j-th column.

Note: since $\det(\mathbf{A}) = \det\left(\mathbf{A}^{\top}\right)$, it is sufficient to see the 2nd part.

Laplace expansion: proof idea

1. By the linearity of det along the j-th column $[\mathbf{a}_j = \sum_{i=1}^{n} a_{i,j}\mathbf{e}_j]$:

$$|\mathbf{A}| = \sum_{i \in [n]} a_{i,j} \left|[\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \mathbf{e}_i, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n]\right|.$$

2. For fixed i, by pushing $\mathbf{e}_i$ to the right and the i-th row to the bottom, it is enough to compute:

$$\left| \begin{bmatrix} \mathbf{A}_{i,j} & \mathbf{0}_{n-1} \\ \mathbf{a}^\top & 1 \end{bmatrix} \right| = |\mathbf{A}_{i,j}|.$$

Laplace expansion: to step-1

Linearity of determinant

Let $\mathbf{a}_1, \ldots, \mathbf{a}_{p-1}, \mathbf{a}_{p+1}, \ldots, \mathbf{a}_n \in \mathbb{R}^n$. Then the mappings $f_{C_p}, f_{R_p} : \mathbb{R}^n \to \mathbb{R}$,

$$f_{C_p}(\mathbf{x}) := \left|[\mathbf{a}_1, \ldots, \mathbf{a}_{p-1}, \mathbf{x}, \mathbf{a}_{p+1}, \ldots, \mathbf{a}_n]\right|,$$
$$f_{R_p}(\mathbf{x}) := \left|\left[\mathbf{a}_1^\top; \ldots; \mathbf{a}_{p-1}^\top; \mathbf{x}^\top; \mathbf{a}_{p+1}^\top; \ldots; \mathbf{a}_n^\top\right]\right|$$

are linear.

Notes:

- Naming: C_p = swaps out column p; R_p swaps out row p.
- Since $\det(\mathbf{A}) = \det\left(\mathbf{A}^\top\right)$, it is sufficient to show the 2nd.

Laplace expansion: to step-1 (proof); $\mathbf{a}_i = [a_{i,j}]_{j=1}^n \in \mathbb{R}^n$

$$f_{R_p}(c\mathbf{x} + d\mathbf{y}) = \left\| \begin{bmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_{p-1}^\top \\ (c\mathbf{x} + d\mathbf{y})^\top \\ \mathbf{a}_{p+1}^\top \\ \vdots \\ \mathbf{a}_n^\top \end{bmatrix} \right\| = \left\| \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{p-1,1} & \cdots & a_{p-1,n} \\ cx_1 + dy_1 & \cdots & cx_n + dy_n \\ a_{p+1,1} & \cdots & a_{p+1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix} \right|$$

$$= \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{1,\sigma(1)} \cdots a_{p-1,\sigma(p-1)} \left(cx_{\sigma(p)} + dy_{\sigma(p)} \right) a_{p+1,\sigma(p+1)} \cdots a_{n,\sigma(n)}$$

$$= c \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{1,\sigma(1)} \cdots a_{p-1,\sigma(p-1)} x_{\sigma(p)} a_{p+1,\sigma(p+1)} \cdots a_{n,\sigma(n)}$$

$$+ d \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{1,\sigma(1)} \cdots a_{p-1,\sigma(p-1)} y_{\sigma(p)} a_{p+1,\sigma(p+1)} \cdots a_{n,\sigma(n)}$$

$$= c \left| \left[\mathbf{a}_1^\top; \ldots; \mathbf{a}_{p-1}^\top; \mathbf{x}^\top; \mathbf{a}_{p+1}^\top; \ldots; \mathbf{a}_n^\top \right] \right| + d \left| \left[\mathbf{a}_1^\top; \ldots; \mathbf{a}_{p-1}^\top; \mathbf{y}^\top; \mathbf{a}_{p+1}^\top; \ldots; \mathbf{a}_n^\top \right] \right|$$

$$= c f_{R_p}(\mathbf{x}) + d f_{R_p}(\mathbf{y}). \quad [(c, 0), (1, 1): \text{classical setting}]$$

Laplace expansion: to step-2 (proof)

Simplified Laplace expansion

Let $\mathbf{A} \in \mathbb{R}^{(n-1)\times(n-1)}$ and $\mathbf{a} \in \mathbb{R}^{n-1}$. Then

$$\left|\begin{bmatrix} \mathbf{A} & \mathbf{0}_{n-1} \\ \mathbf{a}^\top & 1 \end{bmatrix}\right| = |\mathbf{A}|.$$

Indeed, let

$$[a_{i,j}]_{i,j=1}^{n} := \begin{bmatrix} \mathbf{A} & \mathbf{0}_{n-1} \\ \mathbf{a}^\top & 1 \end{bmatrix} \in \mathbb{R}^{n\times n}.$$

For $\sigma \in S_{n-1}$, take $\sigma^* \in S_n$ for which

1. $\sigma(i) = \sigma^*(i)$ $(i \in [n-1])$,
2. $\sigma^*(n) = n$.

Then $\operatorname{sign}(\sigma) = \operatorname{sign}(\sigma^*)$ [we don't prove it], and

$$S_n^* := \{\sigma^* : \sigma \in S_{n-1}\} = \{\pi \in S_n : \pi(n) = n\}.$$

Laplace expansion: to step-2 (proof)

With these notations,

$$\left|\begin{bmatrix} \mathbf{A} & \mathbf{0}_{n-1} \\ \mathbf{a}^\top & 1 \end{bmatrix}\right| \overset{(a)}{=} \sum_{\pi \in S_n} \operatorname{sign}(\pi) a_{1,\pi(1)} \cdots a_{n-1,\pi(n-1)} a_{n,\pi(n)}$$

$$\overset{(b)}{=} \sum_{\pi \in S_n^*} \operatorname{sign}(\pi) a_{1,\pi(1)} \cdots a_{n-1,\pi(n-1)} a_{n,\pi(n)}$$

$$\overset{(c)}{=} \sum_{\sigma \in S_{n-1}} \operatorname{sign}(\sigma^*) a_{1,\sigma^*(1)} \cdots a_{n-1,\sigma^*(n-1)} \underbrace{a_{n,\sigma^*(n)}}_{=a_{n,n}=1}$$

$$\overset{(d)}{=} \sum_{\sigma \in S_{n-1}} \operatorname{sign}(\sigma) a_{1,\sigma(1)} \cdots a_{n-1,\sigma(n-1)} \overset{(e)}{=} |\mathbf{A}|,$$

(a) by def of det, (b) as for $\forall \pi \in S_n \setminus S_n^* \; \exists i \neq n$ s.t. $\pi(i) = n \Rightarrow a_{i,\pi(i)} = a_{i,n} = 0$, (c) reparameterization as $S_n^* \ni \pi = \sigma^* \overset{1:1}{\longleftrightarrow} \sigma \in S_{n-1}$, (d) by the def of σ^* and $\operatorname{sign}(\sigma) = \operatorname{sign}(\sigma^*)$, (e) by def of det.

Laplace expansion (proof): with the ingredients ready

Aim:

$$|\mathbf{A}| = \sum_{i\in[n]} a_{i,j}(-1)^{i+j}|\mathbf{A}_{i,j}|.$$

Step-1: With $\mathbf{A} =: [\mathbf{a}_1, \ldots, \mathbf{a}_n] =: [a_{i,j}]_{i,j=1}^n \in \mathbb{R}^{n\times n}$,

$$\begin{aligned}
|\mathbf{A}| &= |[\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \mathbf{a}_j, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n]| \\
&\overset{(a)}{=} |[\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \sum_{i\in[n]} a_{i,j}\mathbf{e}_j, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n]| \\
&\overset{(b)}{=} \sum_{i\in[n]} a_{i,j} \underbrace{|[\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \mathbf{e}_j, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n]|}_{=(-1)^{i+j}|\mathbf{A}_{i,j}|?}
\end{aligned}$$

(a) holds by $\mathbf{a}_j = \sum_{i\in[n]} a_{i,j}\mathbf{e}_j$, (b) is by the linearity of det.

Laplace expansion (proof – continued)

Step-2:

$$
\begin{aligned}
|[\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \mathbf{e}_j, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n]| &\overset{(a)}{=} (-1)^{n-j} |[\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n, \mathbf{e}_j]| \\
&\overset{(b)}{=} (-1)^{n-j}(-1)^{n-i} \left| \begin{bmatrix} \mathbf{A}_{i,j} & \mathbf{0}_{n-1} \\ \mathbf{a}^\top & 1 \end{bmatrix} \right| \\
&\overset{(c)}{=} (-1)^{i+j} \left| \begin{bmatrix} \mathbf{A}_{i,j} & \mathbf{0}_{n-1} \\ \mathbf{a}^\top & 1 \end{bmatrix} \right| \overset{(d)}{=} (-1)^{i+j} |\mathbf{A}_{i,j}|,
\end{aligned}
$$

(a) $n-j$ column swaps & det change, (b) $n-i$ row swaps, where $\mathbf{a}^\top \in \mathbb{R}^{n-1} = i$-th row of $\mathbf{A}$ after deleting its j-th entry, (c) is by $(-1)^{n-j}(-1)^{n-i} = (-1)^{2n-(i+j)} = (-1)^{-(i+j)} = (-1)^{i+j}$, (d) holds by the simplified Laplace expansion.